Hardware-Aware Real-Time Low-Light Image Enhancement for Improved Lane Detection in Night Driving Scenarios

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Abstract—**Low-light conditions present a major challenge in computer vision, as insufficient illumination reduces visibility and negatively impacts downstream perception tasks such as lane detection.**

This work proposes a lightweight, hardware-aware framework for real-time low-light enhancement aimed at improving lane detection in night driving scenarios.

The proposed system evolved from an initial reflection-based and wavelet-fusion approach into an efficient HSV-domain pipeline that incorporates illumination estimation, adaptive intensity scaling, and sharpening.

The framework was evaluated on low-light image datasets and extended to real-time video processing. It was further integrated with a lane detection module and deployed on a PYNQ-Z2 embedded platform.

Experimental results demonstrate improved scene visibility, more reliable lane extraction, and stable real-time performance of approximately 9–10 FPS under embedded constraints.

Index Terms—**Low-light enhancement, lane detection, embedded vision, PYNQ-Z2, real-time image processing**

I. Introduction

Low-light environments pose significant challenges in computer vision, primarily due to reduced contrast, increased noise, and loss of important visual details.

Images acquired under low illumination conditions typically exhibit degraded visual quality, affecting both human perception and automated vision systems.

These limitations are especially important in applications such as intelligent transportation systems, autonomous driving, and driver assistance technologies.

Among these applications, night driving presents significant difficulty because poor illumination directly affects the visibility of lane markings, road boundaries, pedestrians and obstacles.

Traditional enhancement methods such as Retinex-based approaches and illumination map estimation methods improve visibility but often introduce artifacts or fail under diverse conditions.

Recent computational enhancement approaches achieve strong enhancement quality but often require significant computational resources, limiting their suitability for real-time embedded deployment.

Furthermore, many works evaluate enhancement only visually and do not study its effect on downstream tasks such as lane detection.

This project addresses these limitations through a hardware-aware real-time enhancement framework designed to improve both image visibility and lane detection performance under night driving conditions.

The method evolved from an initial reflection-model and wavelet-fusion-based framework into a lightweight adaptive HSV-domain enhancement pipeline, extended to live video, integrated with lane detection, and deployed on a PYNQ-Z2 embedded platform.

A. Contributions

The main contributions are:

- Development of a lightweight adaptive low-light enhancement framework.
- Experimental evolution from reflection-wavelet fusion to real-time HSV-domain enhancement.
- Integration of enhancement with lane detection.
- Extension from static image enhancement to live video enhancement.
- Embedded implementation on PYNQ-Z2 achieving an average of 9–10 FPS.

II. Related Work

Low-light image enhancement has been widely explored through a variety of classical and recent techniques.

These approaches can generally be grouped into traditional enhancement methods, recent computational enhancement approaches, and task-oriented vision approaches.

A. Classical Low-Light Enhancement

Traditional enhancement methods attempt to improve image visibility through illumination correction, contrast enhancement and structural preservation.

Retinex-based methods model an image as a combination of illumination and reflectance components and estimate illumination for enhancement.

Multiscale Retinex techniques improve contrast and dynamic range but may amplify noise.

Illumination-map-based methods such as LIME estimate illumination distributions to enhance darker regions while preserving structure.

Wavelet-based fusion methods use frequency decomposition to preserve edges and fine details.

The initial phase of this project adopted a reflection-model and wavelet-fusion-based approach as the baseline framework.

B. Recent Computational Enhancement Approaches

Recent low-light enhancement approaches have introduced advanced data-driven and model-driven techniques for improving illumination restoration and detail preservation.

Methods such as Zero-DCE use adaptive curve estimation for brightness correction, while Diff-Retinex combines decomposition-based enhancement with modern restoration strategies.

Illumination-aware gamma correction approaches adaptively adjust brightness while attempting to preserve structural details.

These approaches have demonstrated strong enhancement quality, but many involve increased computational complexity, making real-time embedded deployment challenging.

This motivates the use of lightweight enhancement strategies that prioritize computational efficiency while maintaining acceptable visual improvement.

C. Image to Video Enhancement

Most enhancement algorithms are designed for static images.

When extended to video, temporal consistency becomes important to prevent flickering, frame-to-frame instability, and illumination fluctuations.

In addition to visual consistency, video enhancement introduces low-latency constraints, since each frame must be processed within a limited time budget to support real-time operation.

High computational delay can lead to increased latency, frame drops, and degraded responsiveness, particularly in embedded vision systems.

Recent video enhancement approaches therefore focus not only on maintaining temporal consistency, but also on reducing processing latency through lightweight frame-level operations and computationally efficient pipelines.

These considerations motivated the extension of the proposed method from image enhancement to live video enhancement, where both visual stability and low-latency performance were treated as important design requirements.

D. Lane Detection Approaches

Classical lane detection methods rely on:

- Edge detection
- Region-of-interest masking
- Hough transform
- Geometric line fitting

These approaches are computationally efficient but strongly depend on good edge visibility.

Under low-light conditions, weak edges often lead to poor lane extraction.

This creates a need for enhancement as a preprocessing stage.

E. Research Gap

Based on literature, three major gaps are identified:

- Most methods focus only on enhancement quality.
- Few works evaluate downstream lane detection performance.
- Many methods are not designed for real-time embedded deployment.

To overcome these limitations, a lightweight enhancement framework is developed and integrated with a lane detection pipeline..

III. Problem Statement and Objectives

Road scenes captured under low-light conditions often exhibit reduced visibility, loss of structural clarity, and weakened edge information.

Such degradations impact both visual quality and the performance of downstream tasks, including lane detection.

Conventional enhancement methods may improve brightness, but often introduce artifacts, amplify noise, or fail to satisfy real-time constraints.

This motivates the need for a computationally efficient framework.

A. Problem Definition

Given a low-light driving image or live video frame:

- Improve illumination
- Preserve structural details
- Improve lane visibility
- Support real-time embedded execution

B. Design Constraints

- Limited embedded computation
- Low-latency requirement
- CPU-only execution
- Limited hardware acceleration

C. Objectives

1) Develop a lightweight enhancement pipeline.
2) Improve visibility while preserving edges.
3) Integrate enhancement with lane detection.
4) Extend image enhancement to live video.
5) Deploy on PYNQ-Z2.

D. Scope of Work

Scope includes:

- Low-light enhancement
- Live video enhancement
- Lane detection preprocessing
- Embedded evaluation

FPGA programmable logic acceleration is considered future work.

IV. Proposed Methodology

The proposed system was developed in multiple stages.

$$\text{Input} \rightarrow \text{Enhancement} \rightarrow \text{Lane Detection} \rightarrow \text{Output}$$

A. Baseline Enhancement Framework

Initial framework used:

- Reflection model
- Wavelet fusion
- Adaptive enhancement

B. Experimental Evolution

Several models were explored:

- Adaptive brightness models
- Illumination balancing
- Gamma models
- Sharpened enhancement models

A lightweight adaptive HSV-domain model was selected.

C. Final Enhancement Pipeline

1) Illumination Estimation:

$$I(x, y) = \text{Gaussian}(V) \tag{1}$$

2) Adaptive Brightness:

$$c = \left(1 - \frac{\mu_V}{255}\right) k \tag{2}$$

3) Enhancement:

$$V' = \frac{V(255 + c)}{\max(I, V) + c} \tag{3}$$

4) Post Processing:

- Noise reduction
- Sharpening
- Output stabilization

D. Live Video Enhancement

$$c_t = \alpha c_{t-1} + (1 - \alpha)c \tag{4}$$

E. Lane Detection Module

1) Conversion to grayscale representation
2) Gaussian smoothing
3) Edge extraction using Canny operator
4) ROI masking
5) Line detection using Hough transform
6) Slope filtering
7) Lane overlay

F. Overall Algorithm

1) Acquire frame
2) Convert to HSV
3) Estimate illumination
4) Enhance image
5) Post-process
6) Detect lanes
7) Generate output

Fig. 1 illustrates the complete processing pipeline from input image to final lane detection output.

V. Experimental Setup

This section describes the experimental setup used to evaluate the proposed low-light enhancement framework on both image and live video data.

A. Dataset

Experiments were conducted using:

- A subset of 50 representative low-light road images
- Night driving frames captured under challenging illumination
- Live video streams acquired through a webcam

The dataset includes scenes with:

- Weak lane markings
- Low contrast road boundaries
- Noise affected dark regions
- Non-uniform illumination

B. Software Environment

The proposed final system was implemented using:

- Python
- OpenCV
- NumPy
- Jupyter Notebook

C. Hardware Platform

For embedded validation, the system was deployed on:

- PYNQ-Z2 development board
- Zynq-7020 ARM Processing System
- USB camera input

Due to USB-only input availability, processing was performed using the ARM CPU without programmable logic acceleration.

D. Experimental Configuration

For live processing, the following optimizations were used:

- Input resolution reduced to 240×180
- Camera buffer size minimized
- Lightweight sharpening
- Low-complexity lane detection

(a) Original Input
(b) V-Channel
(c) Enhanced Image
(d) Grayscale
(e) Gaussian Blur
(f) Canny Edges
(g) ROI Mask
(h) Hough Lines
(i) Final Output

Fig. 1. Step-by-step pipeline of the proposed system.

E. Evaluation Metrics

Performance evaluation was carried out using:

- Peak Signal-to-Noise Ratio (PSNR)
- Structural Similarity Index (SSIM)
- Absolute Mean Brightness Error (AMBE)
- Lane detection performance
- Real-time frame rate (FPS)

VI. Results and Analysis

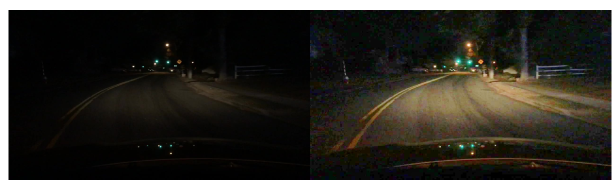
This section presents the performance of the proposed framework in terms of enhancement quality, lane detection improvement, and real-time embedded execution.

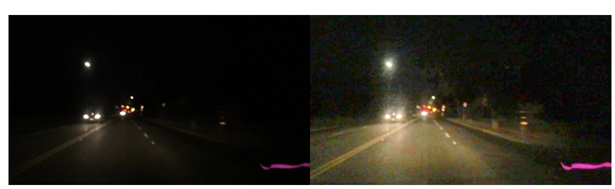
A. Qualitative Enhancement Results

Visual analysis shows that the proposed method improves:

- Overall scene brightness
- Contrast in darker regions
- Visibility of road structures
- Preservation of edge

Figures below show qualitative comparison between original low-light input images and the corresponding enhanced outputs generated by the proposed method.





These results highlight that enhancement quality should be evaluated in the context of downstream vision tasks, where improved structural visibility directly contributes to more reliable lane detection.

B. Quantitative Evaluation

TABLE I
COMPARISON OF ENHANCEMENT METHODS

Method	PSNR	SSIM	AMBE
Adaptive	8.07	0.27	99.60
Domain	20.62	0.77	23.19
Gamma	20.43	0.73	19.40
Proposed	10.84	0.36	69.13

The proposed method prioritizes real-time performance (9–10 FPS) and improved lane detection over peak PSNR and SSIM values.

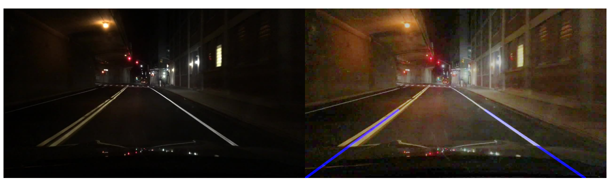
C. Lane Detection Results

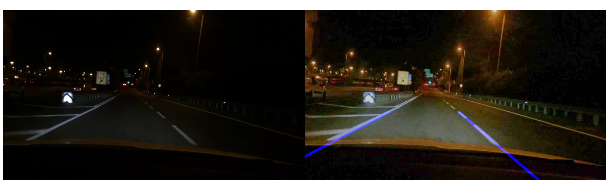
Without enhancement:

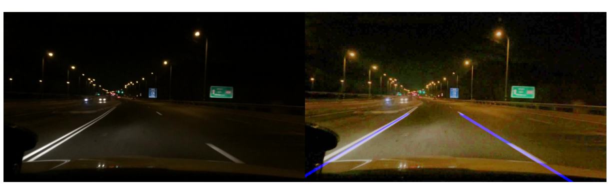
- Weak lane boundaries
- Fragmented edges
- Unstable line detection

With enhancement:

- Improved lane visibility
- Stronger boundary extraction
- More stable lane fitting







D. Live Video Performance

Temporal stability was observed with:

- Reduced brightness fluctuation
- Lower flickering
- Stable enhancement across frames

E. Embedded Performance

The system was deployed on PYNQ-Z2.

Real-time performance achieved:

- Approximate processing speed: 9–10 FPS
- Stable live enhancement output
- Simultaneous lane detection execution

F. Discussion of Results

Results indicate that the proposed framework achieves a favorable trade-off between:

- Enhancement quality
- Lane detection performance
- Computational efficiency
- Embedded feasibility

VII. HARDWARE IMPLEMENTATION ON PYNQ-Z2

To validate embedded feasibility, the proposed enhancement and lane detection framework was deployed on the PYNQ-Z2 development platform.

The implementation was designed to evaluate whether the lightweight enhancement pipeline could support real-time operation under embedded resource constraints.

A. PYNQ-Z2 Platform

The hardware platform used in this work consists of:

- PYNQ-Z2 development board
- Zynq-7020 SoC
- ARM Processing System (PS)
- USB webcam input

The Zynq architecture provides both programmable logic (PL) and an embedded ARM processing system.

In the present implementation, processing was executed using the ARM processing system.

B. System Pipeline

The embedded processing pipeline follows:

$$\text{USB Camera} \rightarrow \text{Frame Capture} \rightarrow \text{Enhancement} \rightarrow \text{Lane Detection} \rightarrow \text{Display Output}$$

C. Implementation Strategy

Due to the available USB-only camera interface, video input was acquired through the processing system, which limited direct use of the HDMI-based programmable logic pipeline.

As a result:

- Enhancement executed on ARM CPU
- Lane detection executed on ARM CPU
- Programmable logic acceleration was not used

D. Optimization for Real-Time Processing

To support low-latency execution, several optimizations were introduced:

- Input resolution reduced to 240×180
- Camera buffer size minimized
- Lightweight filtering used
- Computational complexity reduced

E. Performance

Embedded execution achieved:

- Approximate processing speed: 9–10 FPS
- Stable live enhancement output
- Simultaneous lane detection

Although performance was limited by CPU execution, results demonstrate practical embedded feasibility.

F. Limitations of Current Implementation

The current implementation has the following limitations:

- No programmable logic acceleration
- CPU bottleneck
- Moderate frame rate

These limitations are primarily due to current I/O constraints rather than limitations of the enhancement algorithm.

G. Future FPGA Acceleration Potential

The PYNQ-Z2 architecture supports future acceleration through HDMI input-output based processing.

Future work can offload:

- Illumination estimation
- Enhancement operations
- Edge detection
- Lane extraction

to programmable logic for significantly improved performance.

VIII. Discussion and Limitations

The experimental results demonstrate that the proposed framework achieves a practical balance between enhancement quality, lane detection performance, and embedded feasibility.

The lightweight adaptive enhancement strategy improves low-light visibility while maintaining computational efficiency suitable for real-time execution.

Unlike computationally intensive enhancement methods, the proposed framework prioritizes simplicity and reduced processing overhead, which makes it suitable for embedded deployment.

A. Discussion

Results indicate that enhancement improves not only visual quality but also downstream lane detection performance.

Improved edge visibility in enhanced frames supports more stable lane boundary extraction, showing that enhancement acts as an effective preprocessing stage for perception tasks.

The live video implementation further demonstrates that low-latency processing can be achieved using lightweight frame-level operations.

Although the embedded implementation was limited to CPU execution, real-time performance of approximately 9–10 FPS demonstrates practical feasibility.

B. Limitations

Despite these strengths, the current implementation has several limitations.

- Processing was limited to ARM CPU execution as programmable logic acceleration was not used.
- Frame rate is moderate compared to fully accelerated systems.
- Evaluation was performed on a limited dataset.

C. Trade-Off Analysis

The proposed framework involves a trade-off between:

- Enhancement quality
- Processing speed
- Hardware resource usage

Higher complexity models may improve visual quality, but at the cost of increased latency.

The proposed method intentionally favors computational efficiency for embedded deployment.

IX. Future Work

Several directions exist for extending this work.

A. FPGA Acceleration

A major extension of this work is implementation of the enhancement and lane detection pipeline using programmable logic on the PYNQ-Z2 platform.

Future FPGA acceleration can offload:

- Illumination estimation
- Adaptive enhancement operations
- Edge detection
- Hough-transform-based lane extraction

This can significantly improve frame rate and reduce latency.

B. HDMI-Based Hardware Pipeline

Future implementation using HDMI input-output interfaces can enable direct hardware video processing through the programmable logic pipeline.

This would allow the complete system to move from CPU-based execution to hardware-accelerated real-time operation.

C. Improved Lane Detection

Future work may replace classical lane detection with more robust approaches for challenging road conditions.

Possible extensions include:

- Curved lane modeling
- Robust multi-lane extraction
- Hybrid enhancement-detection pipelines

D. Larger Scale Evaluation

Future experiments can be conducted on larger and more diverse datasets covering wider road and illumination conditions.

This would strengthen generalization analysis.

E. Automotive Embedded Applications

The proposed framework may be extended toward:

- Driver assistance systems
- Embedded automotive vision modules
- Low-light perception preprocessing

X. Conclusion

This work presented a hardware-aware real-time low-light enhancement framework designed to improve visibility and lane detection in night driving scenarios.

The project evolved from an initial reflection-model and wavelet-fusion-based enhancement framework into a lightweight adaptive HSV-domain enhancement pipeline optimized for real-time operation.

The proposed method was extended from static image enhancement to live video enhancement, integrated with a lane detection module, and validated on a PYNQ-Z2 embedded platform.

Experimental results demonstrated improvements in:

- Low-light visibility
- Structural detail preservation
- Lane detection performance
- Real-time embedded feasibility

Embedded implementation achieved approximately 9–10 FPS under CPU-only processing constraints, demonstrating practical feasibility for lightweight embedded vision applications.

Although programmable logic acceleration was not used in the present implementation, the architecture supports future FPGA-based acceleration.

Overall, the results demonstrate that computationally efficient low-light enhancement can improve both visual quality and downstream perception performance while satisfying embedded real-time constraints.

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Appendix A
Project Resources

The complete implementation, dataset, and results for the proposed system are available online.

GitHub Repository:
https://github.com/FAME04/real-time-low-light-lane-detection

Full Project (All Experiments, Models, and Data):
https://drive.google.com/drive/folders/1rgADDbfVgeXslO2C90uwnUoJUyVfa9ZD?usp=sharing

The GitHub repository contains the final implementation and selected results used in this work.

The Google Drive link includes the complete BTP work, including experimental models, intermediate outputs, and additional datasets.